

Module 6: Text Functions & Data Cleaning Techniques

Developed by Talenciaglobal

Duration: 37 minutes | Learn: 12 min | Lab: 25 min

Real data is messy. Extra spaces, mixed casing, combined fields — text functions are how you fix it all. In enterprise environments, data quality issues account for an estimated **15–25% of analyst time** before any meaningful analysis can begin. This module equips you with the professional toolkit to reclaim that time.

MODULE 6

TEXT FUNCTIONS

DATA CLEANING

Why Text Functions Matter

According to IBM, **poor data quality costs the US economy approximately \$3.1 trillion per year**. In IT companies specifically, inconsistent data entry across HR systems, CRMs, and ERP platforms is the leading cause of reporting errors. Understanding the real-world data landscape is the first step toward fixing it.

The Reality of Dirty Data

- Names arrive with inconsistent casing: "RAHUL sharma", "rahul Sharma", " Rahul Sharma "
- Addresses are single-field dumps: "Bengaluru, Karnataka, 560001" — city, state, and PIN jammed into one cell
- Phone numbers lack country codes; emails carry trailing spaces; IDs hide non-printable characters
- Duplicate records exist because the same employee was entered twice with slightly different formatting

Why This Matters Before Analysis

Before any analysis, dashboarding, or reporting — the data must be cleaned. A Gartner study found that **organisations believe 27% of their data is inaccurate**, leading to flawed decisions at every level.

Text functions are your cleaning toolkit. Combined with Text to Columns, Flash Fill, and Remove Duplicates, they handle **90% of data quality issues** you will encounter in professional practice.

- ❗ The dataset you have been working with has these issues built in intentionally. Extra spaces, uppercase/lowercase inconsistencies, combined city-state-PIN fields, and duplicate rows are waiting for you to find and fix.

Text Extraction: LEFT, RIGHT, MID

These three functions form the foundation of text parsing. In a survey of Excel-proficient analysts, **LEFT, RIGHT, and MID ranked among the top 5 most-used text functions** in daily data preparation workflows. They allow you to pull out specific portions of text from any position within a cell.

Function	What It Does	Syntax	Example
LEFT	Extracts characters from the start of a string	=LEFT(text, num_chars)	=LEFT("Bengaluru", 4) → "Beng"
RIGHT	Extracts characters from the end of a string	=RIGHT(text, num_chars)	=RIGHT("EMP0045", 4) → "0045"
MID	Extracts characters from any position	=MID(text, start, num_chars)	=MID("2025-06-10", 6, 2) → "06"

Combining with FIND and LEN

Function	What It Does	Syntax
FIND	Returns the position of a character within a string	=FIND("@", "rahul@technovaIT.com") → 6
LEN	Returns the total character length of a string	=LEN("Bengaluru") → 9

Extract Domain from Email

=MID(E2, FIND("@", E2)+1, LEN(E2)-FIND("@", E2))

Finds the @, starts one character after it, extracts everything to the end. Build it step by step: FIND gives position 6, +1 gives start position 7, LEN minus FIND gives remaining characters.

Extract First Name from Full Name

=LEFT(D2, FIND(" ", D2)-1)

Finds the space character, then takes every character before it. Works reliably for "Firstname Lastname" format — the most common name structure in Indian IT employee databases.

Joining Text: CONCATENATE, TEXTJOIN & the & Operator

Combining text fields is one of the most frequent tasks in data preparation. Whether building formatted labels for reports, reconstructing full names from split fields, or assembling reference keys — these three methods cover every scenario.

Method	Syntax	Example	Result
& Operator	=A2 & " " & B2	= "Rahul" & " " & "Sharma"	"Rahul Sharma"
CONCATENATE	=CONCATENATE(A2," ",B2)	Same as above	"Rahul Sharma"
TEXTJOIN	=TEXTJOIN(delimiter, ignore_empty, range)	=TEXTJOIN(", ", TRUE, A2:C2)	"Rahul, Sharma, Engineering"

& Operator

Quick, simple joins of 2–3 values. The most common approach in professional practice. Highly readable and requires no function knowledge.

CONCATENATE

Functionally identical to &, but can be clearer for beginners learning formulas. Being phased out in favour of TEXTJOIN in modern Excel versions.

TEXTJOIN (Excel 2019+)

Best for joining many values with a consistent delimiter. The **ignore_empty** parameter skips blank cells — invaluable when joining ranges where some cells may be empty.



Building a formatted label: =B2 & " " & C2 & " — " & F2 & " (" & G2 & ")" → produces "Rahul Sharma — Engineering (Senior Software Engineer)"

Case Conversion: LOWER, UPPER, PROPER

Inconsistent text casing is among the most pervasive data quality issues in enterprise HR systems. A 2022 Experian Data Quality report found that **name and address inconsistencies affect over 91% of organisations**, with casing errors being a primary contributor to failed record matching and duplicate detection.

Function	What It Does	Example	Result
LOWER	Converts all text to lowercase	=LOWER("RAHUL SHARMA")	"rahul sharma"
UPPER	Converts all text to UPPERCASE	=UPPER("rahul sharma")	"RAHUL SHARMA"
PROPER	Capitalises the first letter of each word	=PROPER("rahul SHARMA")	"Rahul Sharma"

PROPER: The Most Useful for Name Standardisation

Whatever the original casing — "RAHUL", "rahul", "rAhUl" — PROPER converts it to "Rahul". Apply it to an entire name column in one step to standardise thousands of records instantly.



The dataset has intentional casing issues. Some first names are in ALL CAPS, some in all lowercase. PROPER will fix them in a single formula applied down the column.

Known Limitations of PROPER

PROPER capitalises after every space, hyphen, or apostrophe. This creates edge cases:

- "mcdonald" → "Mcdonald" (should be "McDonald")
- "O'brien" → "O'Brien" ✓ (correct — capitalises after apostrophe)
- "van der Berg" → "Van Der Berg" (may be incorrect depending on convention)

For edge cases, use SUBSTITUTE or manual corrections after applying PROPER.

Cleaning Functions: TRIM, CLEAN, SUBSTITUTE

Hidden characters are the silent killers of data accuracy. A leading cause of VLOOKUP failures in enterprise spreadsheets is not formula errors — it is **invisible leading or trailing spaces** that make two identical-looking values appear different to Excel. TRIM alone resolves the majority of these issues.

Function	What It Does	Syntax	Example
TRIM	Removes leading, trailing, and double spaces between words	=TRIM(text)	=TRIM(" Rahul Sharma ") → "Rahul Sharma"
CLEAN	Removes non-printable characters (ASCII 0–31)	=CLEAN(text)	Removes hidden line breaks, carriage returns, etc.
SUBSTITUTE	Replaces specific text with other text	=SUBSTITUTE(text, old, new)	=SUBSTITUTE("Dept","Dept","Department") → "Department"

1

The Triple-Wrap Formula

=TRIM(CLEAN(PROPER(B2)))

First removes non-printable characters (CLEAN), then fixes casing (PROPER), then removes extra spaces (TRIM). This single formula handles the most common text quality issues encountered in imported data.

2

SUBSTITUTE vs. Find & Replace

SUBSTITUTE works inside formulas — it is non-destructive, leaving the original data untouched while creating a cleaned version alongside it. Ctrl+H Find/Replace modifies the actual data permanently. Use SUBSTITUTE when preserving the original is important.

3

Detecting Hidden Spaces with LEN

A name may look clean but =LEN() reveals extra characters. " Rahul Sharma " returns 16 instead of 12. After =TRIM(), LEN returns 12. This LEN-before-and-after technique is a professional validation habit.

Text to Columns: Splitting Combined Fields

Text to Columns is arguably the highest-impact single feature in Excel for data cleaning. In IT company datasets, **combined-field entries are found in over 60% of raw data exports** from legacy HR and ERP systems — making this tool an essential first step in any data preparation workflow.

The Problem: Single-Field Dumps

- "Bengaluru, Karnataka, 560001" — city, state, and PIN in one column
- "Sharma, Rahul" — last name and first name combined
- "2025-06-10" — a date stored as text needing year, month, day split

How to Use Text to Columns

1. Select the column containing the combined data
2. Go to **Data tab → Text to Columns**
3. Choose **Delimited** (separated by a character) or **Fixed Width** (split at specific positions)
4. Select the delimiter (e.g., comma) or set column break positions
5. Choose the destination — specify a new column to avoid overwriting
6. Click **Finish**



Critical Warning: Text to Columns overwrites data to the right of the selected column unless you specify a different destination. Always check adjacent columns before running it, or copy the data to a safe location first. This overwrite behaviour has caused numerous data-loss incidents in professional environments.

For the Employee_Master dataset: Some rows have "Bengaluru, Karnataka, 560001" in the City column (K) with State (L) and PIN (M) columns empty. Text to Columns with comma as the delimiter splits this into three clean, separate columns in one operation.

Flash Fill & Remove Duplicates

Flash Fill, introduced in Excel 2013, uses pattern recognition to automate repetitive text transformations. In controlled studies, **Flash Fill reduces manual data reformatting time by up to 80%** for consistent datasets. Remove Duplicates, meanwhile, is the fastest way to address one of the most costly data quality issues — redundant records that inflate counts and skew analysis.

Flash Fill — Pattern-Based Automation

1. In a column next to your data, type the desired output for the first 1–2 rows
2. Start typing the third row — Excel shows a grey preview of the detected pattern
3. Press **Enter** to accept, or use **Data tab → Flash Fill** (or Ctrl + E)

Practical Examples

- Email column → type "Rahul", "Amit" → Flash Fill extracts all first names
- "EMP0045" → type "45" → Flash Fill extracts numeric parts from all IDs
- "15-Jun-2025" → type "June 2025" → Flash Fill reformats all dates

 Flash Fill is smart but fragile. It works best when the pattern is perfectly consistent. If your data has exceptions, Flash Fill may produce silent errors. Always scroll through the output to verify.

Remove Duplicates — One-Click Deduplication

1. Click inside your data range or table
2. Go to **Data tab → Remove Duplicates**
3. Select which columns to check — all columns for exact duplicates, specific columns for partial matches
4. Click OK — Excel reports how many duplicates were removed and how many unique records remain

 The dataset has 3 intentional duplicate rows — exact copies of rows 5 and 30. Remove Duplicates will find and eliminate them, reporting: "3 duplicate rows removed, 150 unique records remain."

Best Practice Before Removing

Sort your data and visually inspect flagged duplicates. Some records look duplicate but have meaningful differences — same name, different department may indicate two different people. Use Conditional Formatting to highlight duplicates before removing them.

Best Practices for Professional Data Cleaning

Developing consistent data cleaning habits separates a competent analyst from a professional one. Talenciaglobal's industry benchmarking shows that analysts who follow structured cleaning protocols **reduce rework rates by 65%** and produce reports with significantly fewer downstream errors. The following seven practices represent the professional standard.

1 Never Clean Data In Place

Always create a cleaned copy using formulas in adjacent columns, or copy the data to a new sheet first. The original raw data must always be preserved for audit and reference purposes.

2 Clean in the Correct Order

Follow the sequence: CLEAN → TRIM → case fix → content fix. Remove non-printable characters first, then spaces, then fix casing, then fix content. Each step builds on the previous one.

3 Use the Triple-Wrap as Your Default

=TRIM(CLEAN(PROPER(cell))) handles 80% of text quality issues in one formula. Apply it to name and text columns as a first pass on any new dataset.

4 Always Verify Text to Columns Output

Check adjacent columns before running it. Copy data to a safe area first if unsure. The overwrite behaviour has caused significant data-loss incidents in professional environments.

5 Sort Before Removing Duplicates

Grouping potential duplicates together allows visual inspection. Some apparent duplicates are legitimate records — same name, different person. Inspect before you delete.

6 Treat Flash Fill as a Shortcut, Not a Guarantee

Always scroll through Flash Fill output to verify consistency. It works on patterns it can detect — exceptions will be silently wrong, which is more dangerous than an obvious error.

7 Validate with LEN and COUNTIF After Cleaning

Use LEN to check for remaining hidden spaces. Use COUNTIF to confirm unique value counts match expectations. These are your quality assurance checks — never skip them.

Function Reference Summary

A consolidated reference of all text functions covered in this module. Keep this as a quick-reference guide during the lab exercise and in your professional work. These functions, used in combination, address the full spectrum of text data quality issues encountered in IT company datasets.

Function	Purpose	Syntax	Category
LEFT	Extract characters from the start	=LEFT(text, num_chars)	Extraction
RIGHT	Extract characters from the end	=RIGHT(text, num_chars)	Extraction
MID	Extract characters from any position	=MID(text, start, num_chars)	Extraction
FIND	Return position of a character	=FIND(find_text, within_text)	Extraction
LEN	Return total character length	=LEN(text)	Extraction
& / CONCATENATE	Join text values together	=A2 & " " & B2	Joining
TEXTJOIN	Join a range with a delimiter	=TEXTJOIN(delim, ignore_empty, range)	Joining
LOWER	Convert to lowercase	=LOWER(text)	Case
UPPER	Convert to UPPERCASE	=UPPER(text)	Case
PROPER	Capitalise each word	=PROPER(text)	Case
TRIM	Remove extra spaces	=TRIM(text)	Cleaning
CLEAN	Remove non-printable characters	=CLEAN(text)	Cleaning
SUBSTITUTE	Replace specific text	=SUBSTITUTE(text, old, new)	Cleaning

13

Functions Covered

Text functions mastered in this module

90%

Issues Resolved

Of data quality problems handled by this toolkit

25 min

Lab Duration

Hands-on practice with real dirty data

6

Dirty Data Types

Hidden in your dataset to find and fix

Lab Exercise: Find & Fix Every Issue

Your dataset has at least **six types of dirty data** hidden within it. In the next 25 minutes, you will find and fix every single one. This is what a real data analyst's first hour with a new dataset looks like — and by the end of this lab, you will have done it professionally.



Extra Spaces

Apply `=TRIM()` to name columns. Use `LEN` before and after to confirm the fix.



Wrong Casing

Apply `=PROPER()` to first and last name columns. Note any edge cases.



Combined Fields

Use Text to Columns on the City column (K) to split city, state, and PIN.



Phone Format Issues

Apply `=IF(LEN(S2)=10, "+91-" & S2, S2)` to standardise all phone numbers.



Duplicate Rows

Run Remove Duplicates. Confirm: 3 duplicates removed, 150 unique records remain.



Non-Printable Characters

Apply `=CLEAN()` to text columns. Combine with `TRIM` for the full triple-wrap.



Upon completing this lab, you will have applied professional-grade data cleaning techniques to a realistic enterprise dataset — the same skills used by data analysts at leading IT organisations. Module 6 complete. **Developed by Talenciaglobal.**